MODULE 1.3

UTILITY PROTECTION METHODS

Chapter 1: Underground Utilities · Tier 1 — Field Certified · T.E.A. PROFESSIONAL TRAINING PROGRAM

WHY THIS LESSON EXISTS

You called. You waited. The flags are in the ground. Now what?

A lot of guys think that once the locates are done, they're covered. They're not. Getting the flags in the ground is the beginning of utility protection — not the end. The real work starts when you pick up a shovel or climb into a machine near those marks.

In 17 years as a superintendent, I watched crews with proper locates still hit utilities. Not because they skipped the call — because they didn't know how to work around what was marked. There's a difference between knowing a utility is there and actually protecting it during excavation.

This lesson covers the methods, tools, and field practices that keep utilities intact from the moment you break ground to the moment you backfill.

THE PROTECTION HIERARCHY

Every utility protection decision follows a three-level hierarchy. Work through each level in order before touching the ground:

LEVEL	ACTION	WHAT IT MEANS
1 — Avoid	Design around it	Can you move the trench? Can you bore under it? Elimination is always the best protection.
2 — Expose	Hand dig to confirm	If you can't avoid it, locate it precisely. Hand dig until you can see it, touch it, and measure it.
3 — Protect	Physically support it	If you must work beside or below it, physically support and protect it during all excavation activity.

THE 6 UTILITY PROTECTION METHODS

METHOD 1 — Hand Digging (Potholing)

Hand digging is the first line of defense when working within the tolerance zone of any marked utility. No machine touches the ground until the utility has been manually located and exposed.

When to use:

Any time you are within 18 inches (45 cm) of a locate mark. No exceptions.

How to execute:

- Switch from machine to hand tools at the tolerance zone boundary
- Use a round-point shovel or spade — never a pick or mattock near utilities
- Dig carefully, exposing the utility along its length
- Measure actual depth and horizontal position — note any difference from the marked location
- Flag the exposed utility before resuming machine work
- Maintain visual contact with the exposed utility throughout nearby excavation

FIELD NOTE

Locate marks show WHERE a utility is horizontally — they don't tell you depth. I've uncovered utilities 18 inches shallower than the stated depth. Always treat depth as unknown until you've hand-dug and confirmed it yourself.

METHOD 2 — Hydrovac Excavation

Hydrovac (vacuum excavation) uses pressurized water to break up soil and a vacuum to remove it. It is the safest mechanical method for exposing buried utilities without risk of direct contact.

When to use:

High-density utility corridors, complex crossing points, soft utilities (plastic gas, fiber conduit), situations where hand digging is too slow or impractical.

How to execute:

- Position the hydrovac truck with clear access to the excavation zone
- Use water pressure appropriate to soil type — softer soils require lower pressure
- Work systematically along the utility corridor, not randomly
- Vacuum spoils into the truck immediately — don't pile wet spoil near the excavation
- Expose a minimum 12-inch clearance on all sides of the utility
- Document depth and condition of exposed utilities before proceeding

FIELD NOTE

Hydrovac is increasingly required by municipalities and utility owners for any work within a set distance of high-pressure gas or high-voltage electrical. Check your contract and local requirements before assuming hand digging is sufficient.

METHOD 3 — Mechanical Standoff Distance

When using excavating equipment near marked utilities, operators must maintain a safe standoff distance — the minimum clearance between the machine's bucket and the nearest utility mark.

When to use:

All machine excavation in the vicinity of locate marks, outside the hand-dig tolerance zone.

How to execute:

- Know your machine's bucket reach and swing radius before starting
- Mark the standoff boundary clearly with stakes or paint before the operator begins
- Operator must stop machine travel when the bucket edge reaches the standoff line
- Spotter on the ground maintains visual on both the machine and the locate flags at all times
- If locate marks are disturbed during machine work, stop and re-establish marks before continuing
- Never allow the machine bucket to drag parallel to a utility — approach perpendicularly if possible

FIELD NOTE

Standard standoff for most utilities is 18 inches minimum. For high-pressure gas and high-voltage electrical, many jurisdictions and utility owners require 24–36 inches or more. Check with the utility owner directly if you're unsure — they would rather answer a question than respond to a strike.

METHOD 4 — Utility Shoring and Physical Support

When excavation requires removing the soil supporting an existing utility, the utility must be physically supported to prevent movement, deflection, or damage from ground settlement.

When to use:

Any excavation that undercuts an existing utility, works parallel beneath it, or removes lateral soil support from it.

How to execute:

- Contact the utility owner before any work that will undercut their infrastructure — they may have specific support requirements
- Use timber cribbing, pipe cradles, or engineered support systems as appropriate for the utility type
- Support must be continuous — no point loads or gaps that create bending stress on the pipe or cable
- Never use the utility itself as a lift point or anchor — even momentary load can cause joint failure

- Inspect supports at the start of each shift — settlement and vibration can shift supports overnight
- Get sign-off from utility owner or engineer before removing temporary supports

FIELD NOTE

Gas and water mains are especially vulnerable to joint separation when unsupported. A joint failure on a pressurized main after backfill — which you may not discover for months — is your liability.

METHOD 5 — Protective Barriers and Marking

Physical barriers, warning tape, and visual markers protect utilities during active construction from being struck by equipment, vehicles, or workers who weren't present during the locate process.

When to use:

Any exposed or shallow utility on an active job site where multiple crews or equipment operators are working.

How to execute:

- Install warning tape (detectable tape where required) above all utilities during backfill
- Use concrete or plastic utility markers at grade changes or surface crossings
- Fence or barricade exposed utilities to prevent accidental contact
- Brief all crew members and subcontractors on utility locations at daily tailgate meetings
- Install 'UTILITY BELOW' signage at excavation edges near exposed infrastructure
- Re-mark locate flags if originals are disturbed or faded — never assume everyone on site knows

FIELD NOTE

On multi-trade job sites, the crew that did the locates is rarely the crew working next to the utility two weeks later. Physical barriers and briefings are your only protection against crews who weren't there when the flags went in.

METHOD 6 — Documentation and Communication

Protecting utilities isn't just a physical task — it's a documentation and communication responsibility. What you write down and communicate protects you legally and protects the next crew on site.

When to use:

Every job, every day. Documentation is not optional.

How to execute:

- Record locate ticket numbers and expiry dates in your daily log
- Photograph utilities before covering — show depth measurement and location relative to fixed points
- Note any discrepancies between marked location and actual location found during hand digging
- Record all utility contacts (utility owner calls, locate requests, responses) with date and time

- Brief incoming shifts on any utility issues discovered during your shift
- Ensure as-built drawings reflect any utility relocations or depth corrections discovered on site

FIELD NOTE

Photographs with a tape measure showing actual utility depth have saved contractors from liability claims years after a job was complete. The photo that takes 30 seconds to shoot can be worth hundreds of thousands of dollars in court.

NEVER DO THIS

Never use a pick, breaker, or compactor within the tolerance zone of a marked utility. Never assume a utility is deep enough to be safe. Never remove locate flags before the utility is backfilled and the job is complete. These three actions cause the majority of utility strikes on properly located sites.

SPECIAL CONSIDERATIONS BY UTILITY TYPE

Electrical (Red)

Call the utility owner before any work within 10 feet of high-voltage lines or cables. De-energize where possible. Never assume a cable is dead — test it. Arc flash from buried cables can be fatal with no physical contact.

Natural Gas (Yellow)

Gas lines must be exposed by hand dig or hydrovac only — no exceptions. Even a small scratch on a plastic gas main can cause a slow leak that fills a confined space. If you smell gas at any point, stop work, evacuate, and call the gas company immediately.

Water Main (Blue)

Water mains are under constant pressure. A strike means an immediate flow that can undercut your excavation and create a collapse hazard. Know the location of the nearest shutoff valve before working near any water main.

Telecom / Fiber (Orange)

Fiber lines carry no electrical hazard but are extremely fragile and expensive to repair. Even indirect pressure from a bucket working nearby can crack a fiber conduit. Hydrovac is strongly recommended for exposure near fiber bundles.

Sewer / Storm (Green)

Gravity sewer and storm lines are not pressurized but are usually concrete or clay — brittle and easily cracked by direct or indirect impact. Older clay pipe is particularly vulnerable. Damage often isn't visible until CCTV inspection.

LESSON SUMMARY

Utility protection isn't one action — it's a system of layered methods applied from the moment you receive your locates to the moment you complete backfill. Remember the hierarchy:

1. Avoid — design around the utility if possible
2. Expose — hand dig or hydrovac to confirm exact location and depth
3. Protect — physically support, barrier, and document everything

> **FROM 25 YEARS IN THE FIELD**
>
> The guys who never hit utilities weren't lucky. They were systematic. They hand dug every time. They photographed everything. They briefed their crews. They treated every locate mark like it was a live grenade. That's not fear — that's professionalism.

KNOWLEDGE CHECK

Answer all 5 questions. A score of 4/5 or higher is required to complete this module.

1. What is the first level of the utility protection hierarchy?

a) Expose the utility by hand digging
✓ b) Avoid — design around the utility if possible ← CORRECT
c) Install physical barriers around the utility
d) Contact the utility owner for specifications

2. What is the typical hand-dig tolerance zone on each side of a locate mark?

a) 6 inches
b) 12 inches
✓ c) 18 inches ← CORRECT
d) 36 inches

3. Which excavation method is considered the safest for exposing utilities mechanically?

a) Mini excavator with an experienced operator
b) Hydraulic breaker
✓ c) Hydrovac (vacuum excavation) ← CORRECT
d) Backhoe with a cleanup bucket

4. When must you contact the utility owner directly before excavating?

a) Only if your locate ticket has expired
b) Only on public road projects
✓ c) When working within 10 feet of high-voltage electrical or undercutting any utility ← CORRECT

d) Only when the utility is more than 5 feet deep

5. What is the most important documentation step when exposing a buried utility?

a) Record the utility color code in your daily log

✓ b) Photograph the utility showing measured depth and location relative to fixed points ← CORRECT

c) Call the one-call center to report the find

d) Update the project schedule to reflect the delay

NEXT LESSON → Module 1.4 — Utility Damage Prevention & Response